

IPL Cricket Analytics Portfolio Project

Professional Data Analyst Portfolio Report

1. Executive Summary

This project delivers end-to-end cricket analytics using IPL match and ball-by-ball data. The objective is to transform raw cricket data into actionable insights through Python, MySQL, SQL, and Power BI.

2. Business Problem

IPL franchises, analysts, broadcasters, and cricket enthusiasts require a centralized analytics solution to understand team performance, player impact, venue characteristics, and match-winning factors. Traditional scorecards do not provide strategic insights needed for decision-making.

3. Business Objectives

- Identify the most successful IPL teams.
- Analyze batting and bowling performance.
- Evaluate venue behavior and scoring patterns.
- Measure toss impact on match outcomes.
- Compare chasing versus defending success.
- Create interactive dashboards for stakeholders.
- Generate business recommendations from data.

4. Stakeholders

- Franchise Management
- Coaching Staff
- Team Analysts
- Tournament Organizers
- Broadcasters
- Cricket Fans

5. Dataset Overview

Datasets Used: 1. Matches Dataset 2. Ball-by-Ball Dataset 3. Players Dataset 4. Teams Dataset
Scale: • 1,100+ Matches • 278,000+ Ball Records • Multiple Seasons • Hundreds of Players

6. Data Dictionary

Match Data: match_id, season, venue, team1, team2, toss_winner, toss_decision, match_winner, player_of_match. Ball Data: match_id, innings, over_number, ball_number, batter, bowler, batter_runs, total_runs, is_wicket, wicket_kind.

7. Technology Stack

Python: • Pandas • NumPy • Matplotlib • Seaborn Database: • MySQL Visualization: • Power BI
Version Control: • Git & GitHub

8. Data Cleaning & Preparation

- Missing value analysis.
- Duplicate record validation.
- Data type corrections.
- Handling conditional null values.
- Preserving wicket-event nulls.
- Data quality verification.

9. Exploratory Data Analysis

Team Analysis: • Total Matches • Wins • Win Percentage Batting Analysis: • Top Run Scorers • Strike Rate • Boundary Analysis Bowling Analysis: • Wickets • Economy Rate • Bowling Strike Rate

Venue Analysis: • Average Scores • Matches Hosted

10. Advanced Analytics

• Orange Cap Analysis • Purple Cap Analysis • Powerplay Analysis • Death Overs Analysis • Head-to-Head Analysis • Team Rankings • Venue Intelligence • Chasing vs Defending Analysis

11. SQL Implementation

Implemented 50+ SQL queries covering: • GROUP BY • HAVING • JOINS • CASE WHEN • Subqueries • CTEs • Window Functions • RANK() • DENSE_RANK() • ROW_NUMBER() Business Queries: • Top Players • Team Rankings • Orange Cap Winners • Purple Cap Winners • Venue Analysis

12. Power BI Dashboard Design

Page 1 – Executive Overview KPIs: • Total Matches • Total Runs • Total Wickets • Total Players
Page 2 – Batting Analytics Page 3 – Bowling Analytics Page 4 – Team Analytics Page 5 – Venue Analytics

13. Key Business Insights

1. Certain teams consistently outperform others. 2. Toss decisions influence match outcomes. 3. Some venues are significantly more batting friendly. 4. A small group of players contributes a large share of runs. 5. Death-over specialists have measurable impact on wins. 6. Bowling economy often differentiates successful teams.

14. Recommendations

• Invest in high-impact finishers. • Prioritize venue-specific strategies. • Improve powerplay bowling performance. • Use toss and venue analytics during match planning. • Monitor player consistency across seasons.

15. Resume Project Description

IPL Cricket Analytics Dashboard | Python, MySQL, SQL, Power BI Analyzed 278K+ ball-by-ball records and 1,100+ IPL matches. Performed data cleaning, feature engineering, advanced SQL analysis, and interactive dashboard development. Generated insights related to team performance, player impact, venue intelligence, and match outcomes.

16. Interview Talking Points

• Explain data cleaning decisions. • Discuss SQL window functions. • Describe Power BI dashboard design. • Explain business insights generated. • Discuss challenges handling large datasets. • Explain how analytics supports decision-making.

17. Future Scope

• Machine Learning Match Winner Prediction • Player Performance Forecasting • Live Match Analytics • Advanced Statistical Modeling • Real-Time Dashboards

18. Conclusion

This project demonstrates complete Data Analytics workflow: Data Collection → Data Cleaning → Python Analysis → SQL Analysis → Dashboard Development → Business Reporting → Insights

Generation. The project showcases practical skills expected from a Data Analyst and is suitable for resume, GitHub portfolio, and interviews.